

# Varenya Sri Mudumba

srivarenya@tamu.edu | (979) 344 8215 | College Station, TX | Portfolio | Scholar | LinkedIn | GitHub

## Education

### Texas A&M University, College Station, TX

Aug 2025 - May 2027

M.S. in Computer Science (Thesis track) - GPA 4.0/4.0

Thesis: streaming-ML YARA-rule generation from malware features.

### National Institute of Technology, Durgapur

Dec 2020 - May 2024

B.Tech in Computer Science and Engineering - GPA 8.84/10.00

## Technical Skills

**Languages:** Python, C/C++, Rust, SQL, Bash.

**ML & LLMs:** PyTorch, LLM fine-tuning (LoRA / DoRA), GRPO / RLVR, RAG, transfer learning.

**Techniques:** knowledge distillation, quantization, multi-GPU training, HPC / Slurm, EvalPlus.

**Focus Areas:** malware analysis, NLP, continual learning, hardware security, adversarial ML, AI security.

## Research Experience

### Research Intern

Jul 2026 - Aug 2026

Corelight

Remote, USA

- Designed a novel self-supervised, multi-agent method (Claude on Amazon Bedrock) that reverse-engineers malware through a 23-tool static-analysis MCP to synthesize per-family YARA detection rules.
- Ran systematic experimentation and transcript analysis across a **~38K-sample corpus** (138 families, 6 file types), raising detection coverage from **40-60% to 80-98%** and cutting cost from **~\$10 to ~\$4** per family run.
- Built the evaluation harness and human-in-the-loop review that grounded each iteration's coverage and cost gains, with strict guardrails isolating the agent from raw malware at every stage of the pipeline.

### Graduate Researcher (M.S. Thesis), Dr. Marcus Botacin's Lab

Feb 2026 - Present

Texas A&M University

College Station, TX

- Investigating streaming algorithms for malware analysis to speed up YARA rule generation.
- Designing and evaluating novel and hybrid detection methods for accuracy and efficiency.
- Benchmarking streaming vs. batch feature extraction to quantify accuracy-latency trade-offs at corpus scale.

### Student Research Assistant, GAIA Group

Dec 2025 - Present

Texas A&M University, Dept. of Soil & Crop Sciences

College Station, TX

- Scaling transfer-learning models over multi-spectral data on multi-GPU HPRC clusters for precision agriculture.
- Building reproducible training and evaluation pipelines for large-scale multi-spectral agricultural datasets.

### Undergraduate Researcher (Advisor: Prof. Bibhash Sen)

2023 - 2024

NIT Durgapur

Durgapur, India

- Devised PSO/GA-driven automatic test-pattern generation for hardware-trojan detection, improving fault sensitivity and coverage over prior heuristic baselines on standard ISCAS benchmark circuits.

## Publications

- "PSO-Driven Test Pattern Generation for Hardware Trojan Detection," *IEEE IATMSI 2024*. DOI.

## Projects

### Python Code-Generation SLM

Jun 2026

- Fine-tuned Qwen2.5-Coder-1.5B with DoRA (**70.7% HumanEval** pass@1); added GRPO (RLVR) with an execution-grounded reward (**MBPP 72.5%**) and a hybrid RAG layer that injects exact API signatures at generation time.
- Shipped 2 models and an evaluation dataset on Hugging Face; trained on multi-GPU HPRC clusters.

### Poisoned Pipelines - Agentic-AI & MCP Threat Model

Apr 2026

- Defined **7 threat classes** for agentic AI coding tools and MCP, mapped to OWASP LLM Top 10 and MITRE ATLAS.
- Mapped **17 disclosed CVEs** with STRIDE analysis and layered, defense-in-depth mitigations across the taxonomy.

## Additional

- Merged 2 PRs into IBM's `qiskit-machine-learning`; peak CodeChef rating 1893, 500+ problems solved.
- Work Authorization:** F-1 OPT (STEM) - up to 36 months, no immediate sponsorship.